

MARRI NIKITHA

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PROFESSIONAL SUMMARY

Electronics and Communication Engineering student with knowledge of Embedded Systems, IoT, Arduino, Python, and Wireless Communication. Skilled in antenna design, problem-solving, and software development fundamentals. Seeking opportunities to apply technical skills in Embedded, IoT, and Electronics domains.

TECHNICAL SKILLS

- Programming Languages: Python, C Programming, Java (Basics)
- Web Technologies: HTML, CSS, JavaScript (Basics)
- ECE & Embedded: Arduino, Embedded Systems, Sensors, IoT Basics
- Tools & Platforms: MATLAB, VS Code
- Soft Skills: Teamwork, Leadership, Communication, Time Management

PROJECTS

Audio Amplifier Design and Implementation

Technologies: Analog Electronics, PCB Design, Amplifier Circuits

- Designed and implemented an audio amplifier circuit to enhance low-level audio signals.
- Tested amplifier performance for sound quality, gain, and signal clarity.
- Improved practical knowledge of electronic components, circuit design, and troubleshooting.

Design and Analysis of High Gain Slotted Circular Patch Antenna

Technologies: HFSS, Microstrip Patch Antenna Design, Electromagnetic Simulation

- Designed a high-gain slotted circular patch antenna for enhanced wireless communication applications.
- Analyzed antenna parameters such as gain, return loss, bandwidth, and radiation pattern using simulation tools.
- Optimized slot dimensions to improve antenna performance and efficiency.

IoT-Based Smart Agriculture Monitoring System

Technologies: IoT, Arduino, Sensors, Embedded Systems, Wireless

Communication

- Developed an IoT-enabled agriculture monitoring system to monitor soil moisture, temperature, and humidity in real time.
- Integrated sensors, Arduino, and wireless modules for automated data collection and remote monitoring.
- Improved irrigation efficiency through data-driven decision making and smart resource management.

UART Design and Verification using Verilog HDL

Technologies: Verilog HDL, RTL Design, UART Protocol, Digital Electronics, Verification

- Designed and implemented a UART transmitter and receiver module using Verilog HDL.
- Created testbenches and performed functional verification to validate reliable serial communication.
- Strengthened expertise in RTL design, digital communication protocols, and hardware verification.

AI-Based Attendance System using Face Recognition

Technologies: Python, OpenCV, Machine Learning, Computer Vision

- Developed an AI-powered attendance system using face recognition for automated attendance tracking.
- Implemented image processing and facial recognition algorithms to identify users with high accuracy.
- Enhanced operational efficiency by reducing manual attendance processes and maintaining digital records.

EDUCATION

B.Tech – Electronics and Communication Engineering (ECE)

Vignan's Institute of Management and Technology for Women

2023 – Present | CGPA: 8.0

CERTIFICATIONS

- Python Basics – HP LIFE
- Advanced Software Engineering Job Simulation – Forage
- Getting Started with Generative AI – IBM SkillsBuild
- Data Science Foundations – IBMSkillsBuild
- Cambridge English Interchange
- PCB Design & Fabricationsss
- VLSI FIRST Delivering The Excellence

ACHIEVEMENTS & ACTIVITIES

- Designed and optimized a high-gain slotted circular patch antenna

for wireless communication applications using electromagnetic simulation tools.

- Demonstrated problem-solving, analytical thinking, and teamwork through antenna design and performance optimization.
- Interested in software development, IoT, and automation technologies.

INTERESTS

Reading | sports| Writing | Technology Exploration